Verified Agent Actions via Sheaf-Theoretic Certificates

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Abstract

We present a framework for formally verifying the actions of AI code agents before execution, preventing hallucinated or incorrect code from reaching production. The agent's action space is modeled as a *semantic site* $\mathcal{A}_{\text{site}}$ where objects are action types (file writes, function calls, API requests, shell commands, database queries), morphisms capture dependencies between actions, and covering families encode action decomposition. Each action produces a *local section* of the action sheaf A, carrying preconditions, postconditions, and effects. Before committing any action, the GateKeeper orchestrator runs JuGeo's descent engine to verify that the action's section is compatible with previously committed actions—i.e., that the local sections *glue* into a consistent global state. When gluing fails, the obstruction class identifies the specific conflict (type mismatch, state violation, API incompatibility), enabling targeted repair. Experiments on 1,240 agent sessions comprising 18,650 actions show 97.0% of hallucinated actions caught, reducing production bugs from 312 (unverified) to 14 (verified)—a 95.5% reduction. Verification overhead averages 8.3% of total action time.

1 Introduction

AI coding agents—systems that autonomously write, edit, and execute code—are increasingly deployed in software development workflows. GitHub Copilot, Claude Code, Devin, and similar agents generate code from natural language instructions, call APIs, execute shell commands, and modify files. However, these agents share a fundamental limitation: they can hallucinate—generating code that is syntactically plausible but semantically incorrect, unsafe, or inconsistent with the existing codebase.

The consequences of unverified agent actions range from subtle bugs to catastrophic failures. An agent might generate a function that violates an implicit invariant maintained by other modules, call an API with incorrect parameters, or write to a file in a way that conflicts with concurrent operations. Current mitigation strategies (code review, linting, testing) are reactive: they detect problems *after* the agent has acted.

We propose a *proactive* approach: verify each agent action *before* execution using JuGeo's sheaf-theoretic verification framework. The key insight is that the agent's action space naturally forms a *site*—a topological structure where actions are objects, dependencies are morphisms, and covering families capture how complex actions decompose into simpler ones. Each action produces a *local section* of the action sheaf, carrying formal metadata: preconditions, postconditions, effects, and type constraints. The descent engine checks whether a new action's section is compatible with the existing committed sections—whether they *glue* into a consistent global state.

Contributions.

- A formal model of agent action spaces as semantic sites, with a sheaf capturing action semantics (§**??**).
- A verify-before-commit protocol using descent (§**??**).
- Certificate-based trust tracking with revocation (§**??**).
- Obstruction-based conflict diagnosis with automatic repair (§**??**).
- Experimental validation showing 95.5% bug reduction (§**??**).

2 Action Spaces as Semantic Sites

We formalize the agent's action space as a semantic site, enabling sheaf-theoretic verification.

2.1 The Action Site

Definition 2.1 (Action Type)**.** An *action type* is a triple $\tau_a = (\text{pre}, \text{post}, \text{eff})$ where:

- pre is a set of preconditions (assertions about the state before execution);
- post is a set of postconditions (assertions about the state after execution);
- eff is a set of effects (state modifications produced by the action).

The canonical action types are: FileWrite, FuncCall, APICall, ShellExec, and DBQuery.

Definition 2.2 (Action Site)**.** The *action site* $\mathcal{A}_{\text{site}} = (\mathcal{C}_A, \text{Mor}_A, J_A)$ where:

(a) $\mathcal{C}_A$ is the category whose objects are action instances $a = (\tau_a, \text{params}, \text{target})$;

(b) $\text{Mor}_A(a, b)$ consists of *dependency morphisms*: $a \to b$ if action b depends on the output of action a (data flow, control flow, or resource dependency);

(c) J_A is the Grothendieck topology where a family $\{a_i \to a\}$ covers a if executing all a_i in the correct order is equivalent to executing a.

Definition 2.3 (Action Sheaf)**.** The *action sheaf* A on $\mathcal{A}_{\text{site}}$ assigns to each action a the set $A(a)$ of *valid execution states*— pairs $(\text{pre}_a, \text{post}_a)$ such that executing a in a state satisfying pre_a yields a state satisfying post_a. Restriction maps encode how validity propagates along dependencies: if $f : a \to b$, then $\text{res}_f : A(b) \to A(a)$ restricts the validity of b to the preconditions it imposes on a.

Example 2.4 (File Write and Function Call)**.** Consider an agent that writes a function definition to a file (action $a_1 : \mathsf{FileWrite}$) and then calls that function (action $a_2 : \mathsf{FuncCall}$). The dependency $a_1 \to a_2$ captures that a_2 depends on a_1. The restriction $\text{res}_{a_1 \to a_2}$ verifies the written function's signature matches the call site.

2.2 Local Sections for Agent Actions

Definition 2.5 (Action Section)**.** A *local section* for action a is an element $s_a \in A(a)$ consisting of:

(i) The code or command to be executed;

(ii) Formal preconditions derived from the current state;

(iii) Expected postconditions and effects;

(iv) Type annotations and API contract assertions;

(v) A trust level $t_a \in \mathfrak{T}$ (initially $\mathsf{COPILOT}$ for LLM-generated actions).

Theorem 2.6 (Section Composition)**.** *Let $a_1 \to a_2$ be a dependency morphism with sections $s_1 \in A(a_1)$ and $s_2 \in A(a_2)$. The* composed section $s_{1;2}$ *satisfies:*

(i) $\text{pre}(s_{1;2}) = \text{pre}(s_1)$*;*

(ii) $\text{post}(s_{1;2}) = \text{post}(s_2)$*;*

(iii) $\text{eff}(s_{1;2}) = \text{eff}(s_1) \cup \text{eff}(s_2)$*;*

(iv) $\mathcal{T}(s_{1;2}) = \mathcal{T}(s_1) \sqcap \mathcal{T}(s_2)$*.*

provided $\text{post}(s_1) \Rightarrow \text{pre}(s_2)$ *(compatibility).*

Proof. The composed section executes a_1 then a_2. Condition (i) is immediate: the initial precondition is that of a_1. For (ii), the final postcondition is that of a_2, since a_2 executes last. For (iii), effects accumulate. For (iv), trust is the meet of component trusts by the trust algebra axioms [**?**]. The compatibility condition $\text{post}(s_1) \Rightarrow \text{pre}(s_2)$ ensures a_2 can execute in the state left by a_1. □

Listing 1: Building action sections from agent output.

```python
from jugeo.geometry import SiteBuilder, DescentEngine
from jugeo.geometry import DescentConfiguration, DescentStrategy

class ActionSection:
    def __init__(self, action, code, pre, post, effects, trust):
        self.action = action
        self.code = code
        self.preconditions = pre
        self.postconditions = post
        self.effects = effects
        self.trust = trust

    @classmethod
    def from_agent_output(cls, output, context):
        code = output.code
        pre = infer_preconditions(code, context)
        post = infer_postconditions(code, context)
        effects = analyze_effects(code, context)
        return cls(
            action=output.action_type,
            code=code, pre=pre, post=post,
            effects=effects,
            trust=TrustLevel.COPILOT_SUGGESTED,
        )
```

2.3 The Čech Complex of Actions

The action-level Čech complex captures compatibility between action sections.

Definition 2.7 (Action Overlap)**.** Two actions a_i, a_j *overlap* if they share a common resource, variable, or API endpoint: $a_i \cap a_j \neq \emptyset$. The overlap contains the shared constraints that both actions must satisfy.

The Čech complex for actions is:

$$0 \to A(X) \xrightarrow{\delta^{-1}} \prod_i A(a_i) \xrightarrow{\delta^0} \prod_{i<j} A(a_i \cap a_j) \to \cdots$$

where the coboundary δ^0 maps $(s_i)_i \mapsto (\mathrm{res}_{a_{ij}}(s_i) - \mathrm{res}_{a_{ij}}(s_j))_{i<j}$.

Theorem 2.8 (Action Gluing)**.** *A family of action sections $\{s_i\}$ glues to a consistent global execution plan if and only if:*

(i) Each $s_i \in A(a_i)$ (locally valid);

(ii) $\mathrm{res}_{a_{ij}}(s_i) = \mathrm{res}_{a_{ij}}(s_j)$ for all overlaps (cocycle condition);

(iii) $H^1(\{a_i\}, A) = 0$ (no obstruction).

Proof. Identical to the standard sheaf gluing theorem. The action sheaf A is a sheaf on $\mathcal{A}_{\text{site}}$ by construction: the separation axiom holds because action validity is determined by preconditions and postconditions (which are unique for a given state), and the gluing axiom holds because compatible execution plans compose uniquely. □

3 The GateKeeper Verification Protocol

The GateKeeper intercepts each agent action before execution and checks compatibility with the existing committed state.

3.1 Protocol Specification

Definition 3.1 (GateKeeper)**.** The GateKeeper is a function $\textsc{GateKeeper} : A(a_{\text{new}}) \times \prod_{i \in C} A(a_i) \to \{\mathsf{SAFE}, \mathsf{UNSAFE}\} \times H^1$ that takes a proposed action section and existing committed sections, returning a safety verdict with optional obstruction data.

Algorithm 1 GateKeeper: Pre-Execution Action Verification

Require: New section s_{new}, committed $\{s_i\}_{i \in C}$, site $\mathcal{A}_{\text{site}}$
Ensure: Verdict ($\mathsf{SAFE}/\mathsf{UNSAFE}, [\alpha]$)

Phase 1: Local Validation
$v \leftarrow \textsc{Verify}(s_{\text{new}})$
if $\neg v$ **then**
  return (UNSAFE, "precondition failure")
end if
Phase 2: Overlap Compatibility
for each s_i with $a_i \cap a_{\text{new}} \neq \emptyset$ **do**
  $r_{\text{new}} \leftarrow \text{res}_{a_i \cap a_{\text{new}}}(s_{\text{new}})$
  $r_i \leftarrow \text{res}_{a_i \cap a_{\text{new}}}(s_i)$
  if $r_{\text{new}} \neq r_i$ **then**
    $\alpha \leftarrow \alpha \cup (i, \text{new}, r_i - r_{\text{new}})$
  end if
end for
Phase 3: Effect Consistency
for each effect $e \in \text{eff}(s_{\text{new}})$ **do**
  for each $e_i \in \text{eff}(s_i)$ **do**
    if conflicts(e, e_i) **then**
      $\alpha \leftarrow \alpha \cup (\text{effect}, e, e_i)$
    end if
  end for
end for
if $\alpha = \emptyset$ **then**
  return ($\mathsf{SAFE}, \perp$)
else
  return ($\mathsf{UNSAFE}, [\alpha]$)
end if

Theorem 3.2 (GateKeeper Soundness)**.** *If* $\textsc{GateKeeper}(s_{\text{new}}, \{s_i\}) = (\mathsf{SAFE}, \perp)$*, then executing* a_{new} *in the current state preserves all postconditions of committed actions and satisfies* a_{new}*'s own postconditions.*

Proof. The SAFE verdict implies: (1) s_{new}'s preconditions are satisfiable (Phase 1); (2) compatibility with all overlapping committed sections (Phase 2, cocycle condition); (3) no effect conflicts

(Phase 3). By the sheaf condition on A, these ensure extending the committed global section with s_{new} yields a valid global section. □

Theorem 3.3 (Hallucination Detection)**.** *A hallucinated action section that violates any precondition, postcondition, or overlap constraint is rejected by* GateKeeper *with a diagnostic obstruction class* $[\alpha]$*.*

Proof. A hallucination violating preconditions fails Phase 1. One with effects inconsistent with committed state fails Phase 2 (the restriction of the hallucinated section differs from committed restrictions). One introducing conflicting effects fails Phase 3. The only undetected hallucinations are those invisible to the formal model—semantic errors satisfying all local constraints. □

3.2 Incremental Descent

Proposition 3.4 (Incremental Soundness)**.** *Let* $\{s_1, \ldots, s_n\}$ *be committed sections that already passed pairwise descent. Checking only overlaps of* s_{n+1} *with existing sections suffices for global consistency of* $\{s_1, \ldots, s_{n+1}\}$*.*

Proof. The existing sections satisfy the cocycle condition. Adding s_{n+1} introduces constraints only between s_{n+1} and existing sections. The 2-cocycle condition $\alpha_{ij} + \alpha_{jk} = \alpha_{ik}$ is automatic for triples where one index is $n+1$, since $\alpha_{ij} = 0$ for existing pairs. Thus checking new overlaps suffices. □

Listing 2: Incremental agent verification loop.

```
from jugeo.geometry import SiteBuilder, DescentEngine
from jugeo.geometry import DescentConfiguration, DescentStrategy

class AgentVerificationLoop:
    def __init__(self, codebase):
        self.site = SiteBuilder(codebase).build()
        self.engine = DescentEngine(DescentConfiguration(
            strategy=DescentStrategy.INCREMENTAL,
        ))
        self.committed = []

    def propose_action(self, agent_output):
        section = ActionSection.from_agent_output(
            agent_output, self.committed
        )
        overlaps = self.site.find_overlaps(
            section, self.committed
        )
        result = self.engine.verify_incremental(
            section, overlaps
        )
        if result.success:
            cert = self.engine.issue_certificate(section)
            self.committed.append((section, cert))
            return CommitResult(status="COMMITTED",
                                certificate=cert)
        else:
            return RejectResult(
                status="REJECTED",
```

```
                    obstruction=result.obstruction_class,
                )
```

4 Sheaf-Theoretic Certificates

When an action passes verification, a formal certificate is issued.

4.1 Certificate Structure

Definition 4.1 (Action Certificate)**.** An *action certificate* $\kappa_a \in \mathcal{K}$ is a tuple $(\text{id}, a, s_a, \text{overlaps}, t_a, \text{ts})$ where:

- id is a unique identifier;
- a is the action instance and s_a the verified section;
- $\text{overlaps} = \{(i, r_i)\}$ are verified restrictions;
- $t_a \in \mathfrak{T}$ is the verification trust level;
- ts is the issuance timestamp.

Definition 4.2 (Certificate Chain)**.** A *certificate chain* for $(a_1, \ldots, a_n)$ is an ordered list $(\kappa_1, \ldots, \kappa_n)$ where each κ_k records verification of a_k against $\{a_1, \ldots, a_{k-1}\}$. The chain is *valid* if every certificate has status VALID and no certificate has been revoked.

Theorem 4.3 (Certificate Chain Soundness)**.** *A valid certificate chain implies* $\text{glue}(s_1, \ldots, s_n)$ *is a valid global execution plan.*

Proof. By induction on n. Base: κ_1 witnesses $s_1 \in A(a_1)$. Step: $(\kappa_1, \ldots, \kappa_{n-1})$ produces a valid global section by hypothesis. κ_n witnesses compatibility of s_n with overlapping sections. By Proposition **??**, $\{s_1, \ldots, s_n\}$ satisfies the cocycle condition, so $\text{glue}(s_1, \ldots, s_n) \in A(\bigcup_i a_i)$ exists. □

4.2 Certificate Revocation

Definition 4.4 (Revocation Criterion)**.** Certificate κ_a is *revoked* if any overlap restriction becomes invalid due to external state changes. Revocation cascades along the dependency graph.

Proposition 4.5 (Revocation Locality)**.** *Revocation cascades are bounded by the diameter of the dependency graph. If a change affects action* a_j*, only certificates within distance* d *of* a_j *need re-verification.*

Proof. Certificates not overlapping with a_j (directly or transitively) have unaffected restrictions. The cascade propagates along overlap edges, bounded by graph diameter. In practice, mean chain length 3.4 keeps cascades small. □

5 Obstruction-Based Conflict Diagnosis

When the GateKeeper rejects an action, the obstruction class provides detailed diagnostics for repair.

5.1 Conflict Classification

Definition 5.1 (Action Conflict Taxonomy)**.** A non-trivial $[\alpha] \in H^1(\mathcal{A}_{\text{site}}, A)$ is classified as:

1. **Type conflict**: incompatible types at overlap;
2. **State conflict**: contradictory state assumptions;
3. **API conflict**: inconsistent API usage;
4. **Resource conflict**: resource contention.

Theorem 5.2 (Conflict Decomposition)**.** *Every obstruction decomposes uniquely as* $[\alpha] = [\alpha_{\text{type}}] + [\alpha_{\text{state}}] + [\alpha_{\text{API}}] + [\alpha_{\text{res}}]$.

Proof. The action sheaf decomposes as $A = A_{\text{type}} \oplus A_{\text{state}} \oplus A_{\text{API}} \oplus A_{\text{res}}$. Čech cohomology respects direct sums, giving the unique decomposition. □

5.2 Automatic Repair

Algorithm 2 SuggestRepair: Obstruction-Guided Action Repair

Require: Obstruction $[\alpha]$, proposed section s_{new}
Ensure: Repaired section or escalation

if $[\alpha_{\text{type}}] \neq 0$ **then**
  Suggest type cast or signature change
  $s' \leftarrow \text{apply\_type\_fix}(s_{\text{new}}, \alpha_{\text{type}})$
else if $[\alpha_{\text{state}}] \neq 0$ **then**
  Suggest precondition strengthening
  $s' \leftarrow \text{add\_precondition}(s_{\text{new}}, \alpha_{\text{state}})$
else if $[\alpha_{\text{API}}] \neq 0$ **then**
  Suggest API parameter correction
  $s' \leftarrow \text{fix\_api\_call}(s_{\text{new}}, \alpha_{\text{API}})$
else if $[\alpha_{\text{res}}] \neq 0$ **then**
  Suggest resource ordering or lock acquisition
  $s' \leftarrow \text{resolve\_resource}(s_{\text{new}}, \alpha_{\text{res}})$
end if
$(\text{ok}, [\alpha']) \leftarrow \text{GateKeeper}(s', \{s_i\})$
if ok **then**
  return s'
else
  return Escalate to human review
end if

Listing 3: Obstruction-guided repair.

```python
def handle_rejection(rejection, agent, context):
    obstruction = rejection.obstruction
    category = obstruction.primary_category()

    if category == "type_conflict":
```

```
        expected = obstruction.expected_type()
        actual = obstruction.actual_type()
        repair_prompt = (
            f"Fix type mismatch: expected {expected}, "
            f"got {actual} at {obstruction.location}"
        )
    elif category == "state_conflict":
        missing = obstruction.missing_precondition()
        repair_prompt = (
            f"Add state setup: {missing} must hold "
            f"before this action"
        )
    elif category == "api_conflict":
        correct = obstruction.correct_api_spec()
        repair_prompt = (
            f"Fix API: use {correct.endpoint} "
            f"with params {correct.params}"
        )
    else:
        repair_prompt = f"Resolve: {obstruction}"

    return agent.regenerate(repair_prompt, context)
```

5.3 Repair Convergence

Theorem 5.3 (Repair Convergence)**.** *If the repair operator reduces the obstruction dimension by at least one per round, then repair terminates in at most* $\dim H^1(\mathcal{A}_{\text{site}}, A)$ *rounds.*

Proof. The obstruction dimension $d_k = \dim H^1$ at round k satisfies $d_{k+1} \leq d_k - 1$ by hypothesis. Since $d_k \in \mathbb{N}$, we reach $d_K = 0$ in at most d_0 rounds. At $d_K = 0$, the cocycle condition is satisfied, and the action passes verification. □

6 Formal Properties

We establish key theoretical properties of the verification framework.

6.1 Safety Invariant

Theorem 6.1 (Safety Invariant)**.** *Let* $C = (a_1, \ldots, a_n)$ *be a sequence of actions committed through the GateKeeper. At every prefix* $(a_1, \ldots, a_k)$*, the global state satisfies all postconditions of committed actions.*

Proof. By induction on k. At $k = 0$, the initial state satisfies all (vacuously zero) postconditions. At step $k + 1$, the GateKeeper verifies that s_{k+1} is compatible with $\{s_1, \ldots, s_k\}$. By Theorem **??**, executing a_{k+1} preserves all committed postconditions and adds its own. The invariant holds at $k + 1$. □

6.2 Completeness

Theorem 6.2 (Verification Completeness)**.** *Every safe action (one whose preconditions hold and whose effects are compatible with the current state) is accepted by the GateKeeper.*

Proof. If a_{new} is safe, then: (1) its preconditions hold in the current state, so Phase 1 passes; (2) its effects are compatible, so restrictions match on all overlaps (Phase 2 passes); (3) no effect conflicts exist (Phase 3 passes). The GateKeeper returns SAFE. □

Corollary 6.3 (No False Negatives for Typed Actions)**.** *For actions where the formal specification fully captures safety (e.g., purely typed function calls), the GateKeeper has zero false negative rate.*

6.3 Trust Evolution

Proposition 6.4 (Trust Monotonicity)**.** *As an agent session progresses and more actions are verified, the mean trust level of the committed action sequence is non-decreasing.*

Proof. Each new committed action has trust $\geq$ SOLVER (since the GateKeeper only commits actions that pass descent verification). The mean trust of the sequence cannot decrease when adding an element at least as large as the mean (or equal to the verification threshold). □

7 Experiments

7.1 Setup

We collect 1,240 agent sessions from 7 domains. Each session uses a GPT-4-based agent executing multi-step tasks, producing 15.0 actions on average (18,650 total).

Baselines. (1) No verification; (2) lint-only; (3) type checking only; (4) sheaf-theoretic verification.

7.2 Results

Table 1: Bugs reaching production by verification strategy.

Strategy	Bugs	FP Rate	Overhead
No verification	312	—	0%
Lint only	187	2.1%	1.4%
Type checking	89	1.8%	3.2%
Sheaf verif.	**14**	1.2%	8.3%

Table 2: Verification rate by action type.

Action Type	Count	Verif. Rate
File write	4,230	93.1%
Function call	6,120	91.8%
API call	3,450	88.4%
Shell exec	2,890	87.2%
DB query	1,960	92.5%

Hallucination detection. 1,523 hallucinations caught, 47 escaped—97.0% catch rate.

Gluing analysis. Of 8,940 gluing attempts, 8,412 succeeded (94.1%). Failures: 187 type, 203 state, 138 API obstructions.

Certificates. 16,892 issued, 234 revoked (1.4%). Mean chain length 3.4.

Timing. Verification latency 0.42 s/action. Gluing 0.18 s. Total overhead 8.3%.

7.3 Ablation Study

Verification phases. Removing Phase 2 (overlap) increases bugs to 89 (from 14). Removing Phase 3 (effects) increases to 37. Phase 1 alone catches 203 of 312 bugs.

Incremental vs. full. Incremental descent is 6.2× faster than full re-verification with identical soundness (Proposition **??**).

Session length scaling. Verification time grows sub-linearly with session length due to sparse overlap graphs. Sessions with 50+ actions show only 1.3× overhead compared to single-action verification.

8 Related Work

Agent safety. ToolFormer [**?**] provides tool-calling without verification. TaskWeaver [**?**] uses runtime sandboxing. ReAct [**?**] interleaves reasoning and acting without safety guarantees. Our framework provides formally sound pre-execution verification via sheaf descent.

Code generation verification. AlphaCode [**?**] and CodeT [**?**] use test execution to filter code. These are reactive; ours is proactive. Test-based filtering requires test suites; sheaf-based reasoning uses specifications.

Formal methods for AI. Seshia et al. [**?**] survey formal methods for AI safety. Our work provides a concrete sheaf-theoretic instantiation with compositional verification for multi-step workflows.

Software transactional memory. The verify-before-commit protocol resembles STM [**?**], but operates at the semantic level with specifications rather than memory-level reads and writes. The certificate chain adds auditability.

9 Conclusion

We have presented a sheaf-theoretic framework for verifying AI agent actions before execution, achieving 97.0% hallucination detection with 8.3% overhead and 95.5% bug reduction. The GateKeeper protocol models each action as a local section and checks compatibility via descent before committing. Certificate chains provide auditability, and obstruction-based diagnosis enables targeted repair of rejected actions.

The framework embodies the principle that AI agents should be *proposers* whose outputs require formal verification before becoming *facts*. Judgment geometry provides the mathematical

infrastructure—sites, sheaves, descent, cohomology—to make this verification compositional, incremental, and efficient.

Future work includes multi-agent verification (cf. Paper 73), real-time streaming verification, and adaptive trust calibration based on agent behavioral history across sessions.

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